Homework 7 — changed numbers (extra practice)

Module 7b - PK-PD Relationship · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A 70-kg volunteer received a single IV bolus dose of an investigational drug. The drug was administered at a dose of 12 mg/kg. The drug has an apparent volume of distribution that is 0.25 L/kg and an elimination half-life approximately 5 hours.

1a What is the AUC (area under the curve) following the IV bolus dose?

1b If the minimum effective concentration (MEC) is 4 mg/L, what is the duration of activity of this drug in this volunteer?

1c What is the duration of activity if the dose were 6 mg/kg?

1d What dose would provide a duration of action of 10 hours?

2. A 72-kg, 31-year-old female is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 250 mg, her steady state plasma concentration was 6.4 mg/L. The daily dose was increased to 350 mg which resulted in

a steady-state plasma concentration of 17.6 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

2a What is this patient's KM?

2b What is the Vmax in this patient?

2c What is the daily dose rate of phenytoin that should achieve a steady state phenytoin plasma concentration of 14 mg/L in this patient? (Round to the nearest ten)

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 14 mg/L? What is the half-life?

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 14 mg/L? What is the clearance?

2e What is the expected steady state concentration if this patient were to take 300 mg/day?